NLP

23-06-26

Attention Mechanism:

i am flying a kite

↓

i am fly #ing a kite

↓ embedding layer

e_0 e_1 e_2 e_3 e_4 e_5

\+ p_0 p_1 p_2 p_3 p_4 p_5

↓

$\bar{e}_0$ $\bar{e}_1$ $\bar{e}_2$ $\bar{e}_3$ $\bar{e}_4$ $\bar{e}_5$ — position aware embeddings

↓ attention

e'_0 e'_1 e'_2 e'_3 e'_4 e'_5 — context aware embeddings

Steps in attention mechanism:

for an individual embedding $\bar{e}_3$

1) Pass the position aware embedding through Query weight matrix and get Query vector

$$q_3 = \bar{e}_3 \cdot W_q$$

$1 \times d \quad 1 \times d \quad d \times d$

W_q - Query weight matrix
- randomly initialized at start
- its weights are learnable model parameters

q_3 - query vector/embedding

2) Pass all position aware embeddings through Key Weight matrix and get Key vector matrix

$$\bar{e}_0 \times W_K = K_0$$
$$\bar{e}_1 \times W_K = K_1$$
$$\bar{e}_2 \times W_K = K_2$$
$$\bar{e}_3 \times W_K = K_3$$
$$\bar{e}_4 \times W_K = K_4$$
$$\bar{e}_5 \times W_K = K_5$$

$$K = \bar{e} \times W_K$$
$$6 \times d \quad 6 \times d \quad d \times d$$

W_K - key weight matrix
- its learnable
- randomly initialised at start

K - key matrix

3) compute dot products of query vector and all vectors in key matrix

q_3 - query vector

K - key matrix

dot products : $q_3 . k_0 = d_0$
$q_3 . k_1 = d_1$
$q_3 . k_2 = d_2$
$q_3 . k_3 = d_3$
$q_3 . k_4 = d_4$
$q_3 . k_5 = d_5$
} scaler values

$q_3 = [1 \quad 2 \quad 3]$
$k_0 = [1 \quad 0 \quad 1]$
} $q_3 . k_0 = 1\times1 + 2\times0 + 3\times1 = 4$

dot products

$D = q_3 K^T$

$1\times6 \quad 1\times d \quad d\times6$

$D = [d_0 \quad d_1 \quad d_2 \quad d_3 \quad d_4 \quad d_5]$

scale the dot products using $\sqrt{d}$

d - dim of query vector

$$D = \frac{D}{\sqrt{d}}$$

$$d_0 = \frac{q_3 . k_0}{\sqrt{d}} \quad d_1 = \frac{q_3 . k_1}{\sqrt{d}} \quad d_2 = \frac{q_3 . k_2}{\sqrt{d}} \quad d_3 = \frac{q_3 . k_3}{\sqrt{d}}$$

$$d_4 = \frac{q_3 . k_4}{\sqrt{d}} \quad d_5 = \frac{q_3 . k_5}{\sqrt{d}}$$

4) Pass the scaled dot products through softmax activation function

$d_0 \quad d_1 \quad d_2 \quad d_3 \quad d_4 \quad d_5$

$\downarrow$ softmax

$a_0 \quad a_1 \quad a_2 \quad a_3 \quad a_4 \quad a_5$

$$a_i = \frac{e^{d_i}}{\sum_{i=0}^{5} e^{d_i}}$$

$A = [a_0 \quad a_1 \quad a_2 \quad a_3 \quad a_4]$

Output of softmax – a_0 a_1 a_2 ... a_5 are called attention scores

5) Pass all position aware embeddings $\bar{e}_0$ $\bar{e}_1$ ... $\bar{e}_5$ through Value Weight matrix and get Value Matrix

$$V = \bar{e} W_V$$

$6 \times d \quad 6 \times d \quad d \times d$

6) Get context aware embedding using attention scores and Value Matrix

$$e_3' = A V$$

$1 \times d \quad 1 \times 6 \quad 6 \times d$

$A = [a_0 \quad a_1 \quad a_2 \quad a_3 \quad a_4 \quad a_5]$

$a_0 + a_1 + a_2 + a_3 + a_4 + a_5 = 1$

these are computed for q_3

now if a_5 value is higher ≈ 0.35

this means that tokens t_3 and t_5 have a strong relationship between them

the context aware embedding e_3' will be highly influenced by t_5

$\bar{e}_3 \xrightarrow{W_Q} q_3 \qquad \bar{e} \xrightarrow{W_K} K \qquad \bar{e} \xrightarrow{W_V} V$

$$A = \text{softmax}\left(\frac{q_3 \cdot K^T}{\sqrt{d}}\right)$$

$$e_3' = A \cdot V$$

$$e_3' = \text{softmax}\left(\frac{q_3 \cdot K^T}{\sqrt{d}}\right) V$$

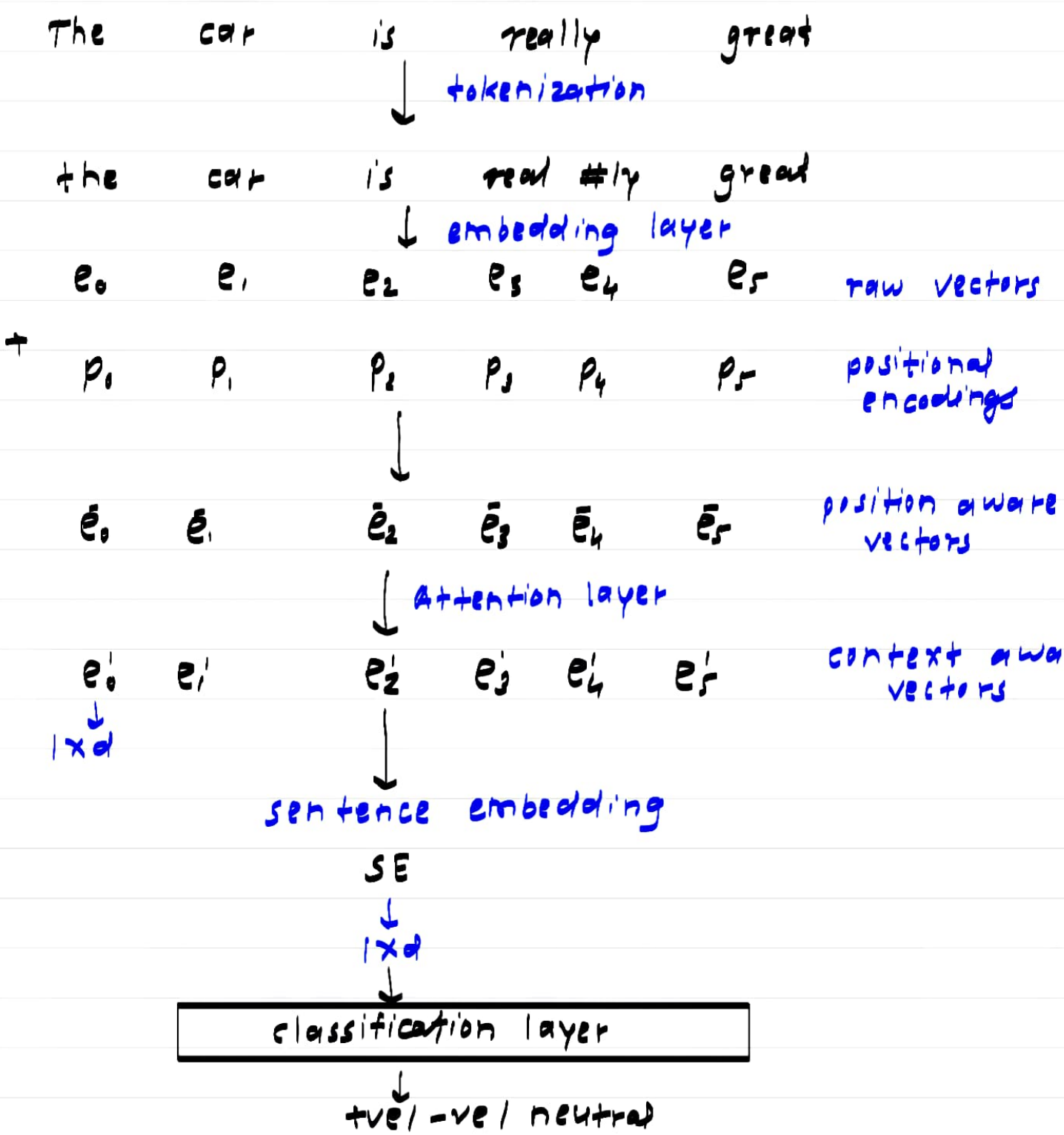
for all position aware embeddings

$\bar{e} \xrightarrow{W_Q} Q$ — query matrix

$$\boxed{e' = \text{softmax}\left(\frac{Q K^T}{\sqrt{d}}\right) V}$$

matrix form of attention mechanism

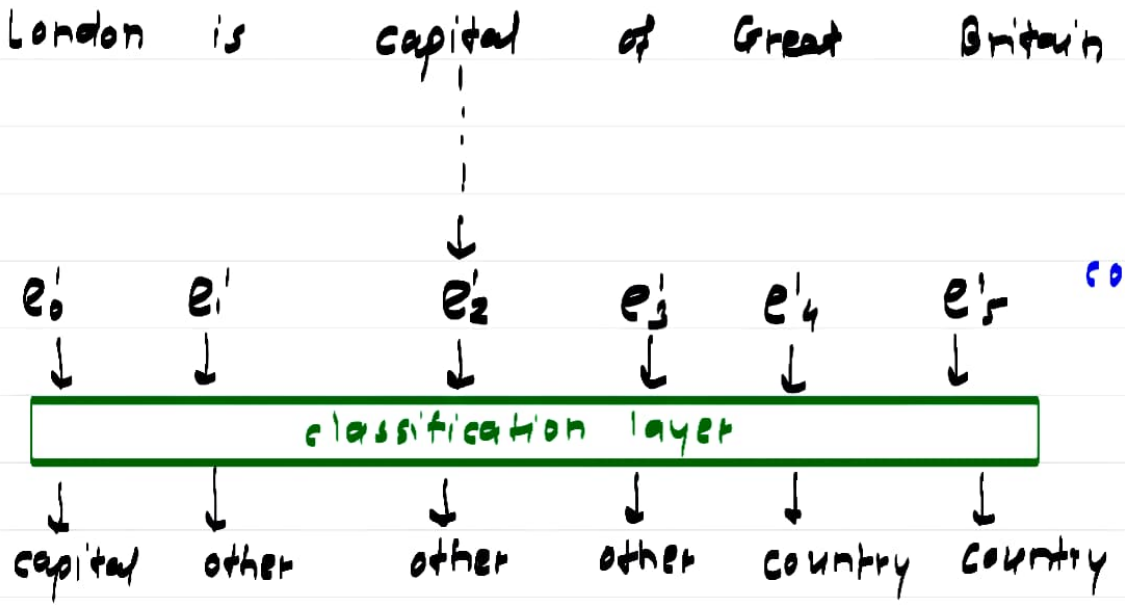
Text classification using Transformer :



lets say d = 5



Token classification using Transformer:



In attention mechanism long distance context is not lost

$$a_{200} = \text{softmax}\left(\frac{q_2 \cdot k_{200}}{\sqrt{d}}\right)$$



Transformer models:

1) Encoder only transformers
   - text classification / token classification
   - used in RAG systems

2) Encoder - Decoder transformers
   - sequence to sequence tasks
   - QnA, translation

3) Decoder only transformer

- sequence to sequence task
- Unified all types of NLP tasks
- Generalised NLP models
- most of the modern LLMs are of this type
  - GPT
  - Claude
  - Gemini

encoder only Decoder only

self attention

Encoder + Decoder

self attention + cross attention